Course Code: CSE445 | **Section:** 06 | **Project Group:** 7

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Real-World Traditional ML Object Tracker

***Abstract*—Multi-object tracking (MOT) in unconstrained video environments remains a central challenge in computer vision, requiring robust detection and state-estimation under conditions such as occlusion, camera motion, cluttered backgrounds, and varying lighting. This paper presents a complete, production-quality MOT pipeline built exclusively from classical signal-processing and machine-learning components—Gaussian Mixture background subtraction (MOG2/GMG), Lucas–Kanade optical-flow camera stabilisation, Kalman filtering, the Hungarian assignment algorithm, and HSV colour-histogram re-identification—augmented with YOLOv8 neural detection as an optional high-accuracy module. A structured three-way dataset split (train / validation / test) across ten heterogeneous video sequences drives a grid search over 40 hyperparameter combinations, enabling systematic selection of the best detector and tracker configuration per scene type. Evaluation with standard metrics (Precision, Recall, F1-score, MOTA, ID Switches, and centroid MAE) demonstrates that no single algorithm dominates all scenarios; instead, detector selection must be coupled to scene characteristics. YOLO consistently outperforms background-subtraction methods in dynamic or multi-class environments, while MOG2 provides a computationally efficient baseline for static-camera, single-class sequences. These findings validate the hypothesis that principled algorithm selection, guided by video diagnostics and systematic hyperparameter tuning, yields substantially better tracking accuracy than any fixed pipeline. The full pipeline is implemented as a Google-Colab-compatible Jupyter Notebook and operates end-to-end on user-uploaded videos with no manual labelling required.**

***Index Terms*—multi-object tracking, background subtraction, Kalman filter, Hungarian algorithm, YOLOv8, MOG2, GMG, grid search, MOTA, re-identification, optical flow, camera stabilisation**

I. Introduction

Motion tracking is the computational process of localising one or more objects across successive video frames and maintaining a consistent identity label for each throughout the sequence [1]. Its applications span a diverse and critical range of domains: autonomous vehicles rely on tracking pedestrians and adjacent vehicles to plan safe trajectories; surveillance systems use it to detect intrusions and loitering; sports analytics exploit trajectory data to quantify athlete performance; robotics depends on it for real-time obstacle avoidance; and medical imaging uses cell or instrument tracking to guide procedures. Given these high-stakes applications, even marginal improvements in tracking fidelity translate directly into measurable safety or operational gains.

The central difficulty in MOT arises not from any single algorithmic deficiency but from the enormous diversity of visual conditions encountered in real deployments. Fast-moving objects generate motion blur that defeats frame-differencing and classical feature detectors. Densely crowded scenes produce frequent occlusions, forcing a tracker to maintain object identity across periods of complete invisibility. Cluttered or temporally dynamic backgrounds—flickering illumination, waving foliage, reflective surfaces—overwhelm simple statistical background models and generate high false-positive rates. Finally, moving cameras invalidate the fundamental assumption of background subtraction: that the scene background is globally stationary.

Classical approaches address these challenges through a layered pipeline architecture. Background subtraction (e.g., MOG2 [2], GMG [3]) separates foreground blobs from the scene model; morphological post-processing refines blob boundaries; Kalman filtering [4] predicts each track's state across frames in which detections are absent; the Hungarian algorithm [5] solves the combinatorial assignment between predicted track positions and current detections; and re-identification modules recover track continuity after occlusion. Deep-learning detectors such as YOLOv8 [6] replace the detection front-end with a learned feature extractor, providing higher per-frame accuracy at the cost of GPU resources.

Despite the maturity of these individual components, their *joint* behaviour under diverse scene conditions remains under-studied. Practitioners typically fix a single configuration and apply it universally, often yielding suboptimal results. This work addresses that gap with the following core problem statement:

This work evaluates multiple motion tracking approaches under varying scene conditions—encompassing static and moving cameras, single and multiple objects, and clean and cluttered backgrounds—to determine the effectiveness of each approach and to establish principled guidelines for algorithm selection.

The specific contributions of this paper are:

1) A modular, Google-Colab-compatible MOT pipeline integrating MOG2, GMG, and YOLOv8 detectors within a unified Kalman + Hungarian + Re-ID tracking backend, implemented across 12 self-contained notebook cells.
2) A systematic 40-combination hyperparameter grid search (24 MOG2 + 8 GMG + 8 YOLO) evaluated on six training video sequences.
3) A rigorous three-way train/validation/test evaluation protocol, reporting Precision, Recall, F1, MOTA, MAE, and ID Switches.
4) Empirical evidence that detector–scene matching is the dominant factor in tracking performance, with YOLOv8 achieving the highest F1 and MOTA in high-motion sequences and MOG2 remaining competitive in static-background settings.
5) An automated YOLO-based ground-truth annotation pipeline (Cell 4c) with per-video class filtering, eliminating the need for manual bounding-box labelling.

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II. Related Work

A. Background Subtraction Methods

Background subtraction is one of the oldest paradigms in video analysis. Wren et al. introduced Pfinder [10], a single-Gaussian skin-colour model for person tracking. Kaew-TraKulPong and Bowden [11] extended this to a Gaussian Mixture Model (GMM), enabling multiple stable background colours per pixel. Zivkovic [2] further improved GMM with adaptive component selection, yielding the MOG2 implementation available in OpenCV. Godbehere et al. proposed GMG [3], a Bayesian frame-count model that provides sharper initial segmentation but requires a long initialisation period (60 frames in the present implementation).

B. State-Space Tracking

The Kalman filter [4] underpins most practical MOT systems. Its extension to the joint detection-and-tracking paradigm via SORT (Simple Online and Realtime Tracking) [8] demonstrated that Kalman prediction combined with IoU-based Hungarian assignment achieves competitive MOTA with millisecond-level latency. DeepSORT [9] added a deep appearance descriptor for improved re-identification, at the cost of a separate embedding network.

C. Deep-Learning Detectors in Tracking

YOLO [12] demonstrated that single-pass convolutional detection could achieve real-time throughput. YOLOv8 [6] continues this lineage with improved accuracy, anchor-free heads, and a versatile Python API. Integrating YOLO as the detection front-end in tracking pipelines is now standard practice, with the downstream tracker remaining classical (Kalman + Hungarian) in most industrial systems.

D. Positioning of the Present Work

Unlike SORT and DeepSORT, the present system is fully self-contained, avoids deep re-ID networks, and includes a systematic comparison across three detector types driven by an automated grid search. The emphasis on *comparative* evaluation under diverse scene conditions distinguishes this work from single-algorithm evaluations. The use of a fixed random seed (`RANDOM_SEED = 42`) and a structured three-way split ensures full reproducibility.

III. Methodology

A. Detector Modules

1) MOG2 — Mixture-of-Gaussians Background Subtraction: MOG2 models the distribution of each pixel's intensity history as an adaptive mixture of K Gaussian components. For pixel $\mathbf{x}$ at time t:

$$P(\mathbf{x}_t) = \sum_{k=1}^{K} \omega_{k,t}\, \mathcal{N}\big(\mathbf{x}_t;\, \boldsymbol{\mu}_{k,t},\, \Sigma_{k,t}\big) \tag{1}$$

where $\omega_{k,t}$, $\boldsymbol{\mu}_{k,t}$, and $\Sigma_{k,t}$ are the weight, mean, and covariance of the k-th component. A pixel is labelled foreground when it falls below the lowest-ranked background components. MOG2 incorporates shadow detection via an additional grey-world prior and supports an adaptive variance threshold τ_σ, which was a key grid-search parameter in this work (values 30, 50, 70). A history window of 300 frames is used and shadow pixels (mask value 127) are suppressed before morphological post-processing.

Why MOG2? For static-camera sequences with a relatively homogeneous background—such as the overhead pedestrian camera or the store surveillance feed—MOG2 offers an excellent efficiency–accuracy trade-off. Its $\mathcal{O}(K)$ per-pixel update is compatible with real-time processing on CPU hardware, achieving 30–60 FPS in practice.

2) GMG — Bayesian Counting Model: GMG [3] maintains a per-pixel observation count and a Bayesian estimate of the background distribution. During an initialisation phase of 60 frames, it accumulates pixel statistics before declaring any pixel as foreground. Once initialised, its foreground probability map is sharper than MOG2's but it produces no usable output during the warmup period. The pipeline enforces a 65-frame warmup guard for GMG to prevent spurious track initialisation during the burn-in phase.

Why GMG? In sequences with rich background texture that causes MOG2 to produce scattered false positives—such as the game screen (Aim Lab) sequences—GMG's stronger statistical model reduces clutter at the cost of slower initialisation and higher CPU load.

3) YOLOv8 — Single-Stage Neural Object Detector: YOLOv8 [6] divides the input image into an implicit grid and simultaneously regresses bounding boxes and class probabilities in a single forward pass through a deep convolutional backbone. It is trained on COCO (80 classes) and supports arbitrary class filtering at inference time. Detections below a confidence threshold τ_{conf} or below a minimum bounding-box area A_{min} are discarded. In this pipeline, the `yolov8n` (nano) variant is used to balance speed and accuracy. Per-video class filters are applied at inference time: class 0 (person) for dance sequences, class 16 (dog) for the puppy clip.

Why YOLO? Background subtraction inherently fails when the camera moves (optical-flow stabilisation only partially compensates) or when the scene contains multiple, semantically distinct object classes. YOLO's learned feature hierarchy is invariant to these conditions. For the handheld-camera dog and football sequences, YOLO was the only detector capable of isolating the target objects cleanly. Critically, YOLO requires no warmup period, providing valid detections from frame zero.

B. Camera Motion Compensation

A Lucas–Kanade pyramidal optical-flow estimator [7] tracks up to 200 Shi–Tomasi corner features between consecutive frames. Features are detected with quality level 0.01, minimum distance 10 pixels, and block size 5. A partial-affine transformation $\mathbf{H} \in \mathbb{R}^{2\times 3}$ is estimated via RANSAC (reprojection

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threshold 3 pixels) from at least 6 matched features and applied to warp the current frame onto the reference coordinate system:

$$\hat{I}_t = \mathcal{W}(I_t,\, \mathbf{H}_t) \tag{2}$$

This step is activated automatically for videos flagged as handheld-camera sequences in the `VIDEO_MOVING_CAMERA` dictionary. It is disabled for static-camera footage, avoiding unnecessary computation and potential warping artefacts.

C. Kalman Filter State Estimator

Each tracked object maintains an independent six-state Kalman filter with state vector $\mathbf{x} = [x,\, y,\, w,\, h,\, v_x,\, v_y]^\top$, where (x, y) is the top-left corner of the bounding box, (w, h) its width and height, and (v_x, v_y) the constant-velocity components. The state-transition and measurement matrices are:

$$\mathbf{F} = \begin{bmatrix} 1 & 0 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}, \quad \mathbf{H} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \end{bmatrix} \tag{3}$$

The initial state covariance is $\mathbf{P} = 500\,\mathbf{I}_6$. Measurement noise $\mathbf{R} = r\,\mathbf{I}_4$ and process noise $\mathbf{Q}_{4:,\,4:} = q\,\mathbf{I}_2$ were held fixed in the grid search ($r = 10$, $q = 1$), as preliminary experiments showed negligible variation; only `max_disappeared` and `max_distance` had material impact. Object speed is computed as speed $= \sqrt{v_x^2 + v_y^2}$ and stored for diagnostics.

D. Hungarian Assignment

At each frame the cost matrix $\mathbf{C} \in \mathbb{R}^{N_{\text{tracks}} \times M_{\text{dets}}}$ is computed as a convex combination of spatial and appearance costs:

$$C_{ij} = 0.7\,\frac{d_{ij}}{d_{\max}} + 0.3\,(1 - s_{ij}) \tag{4}$$

where d_{ij} is the Euclidean centroid distance between predicted track i and detection j, $d_{\max}$ is the maximum allowable distance (a grid-search parameter), and s_{ij} is the HSV histogram correlation between the track's stored histogram and the detection's current histogram. Assignments where $d_{ij} > d_{\max}$ are rejected even if they satisfy the Hungarian optimum. The Hungarian algorithm [5] solves $\min \sum_{(i,j)\in\mathcal{M}} C_{ij}$ in $\mathcal{O}(n^3)$ time via `scipy.optimize.linear_sum_assignment`.

E. HSV Colour-Histogram Re-Identification

When a track exceeds the `max_disappeared` patience threshold it is moved to a *dormant* buffer (capped at 30 entries to prevent memory growth). Any new, unmatched detection is compared against all dormant tracks using the Bhattacharyya correlation of 18-bin hue $\times$ 16-bin saturation histograms, computed over the bounding-box ROI in HSV colour space. If the best similarity $s^* \geq \tau_{\text{ReID}}$ (default 0.55) the detection is assigned to the dormant track and the track is reactivated. This recovers identity continuity across brief occlusions without the computational overhead of deep embedding networks.

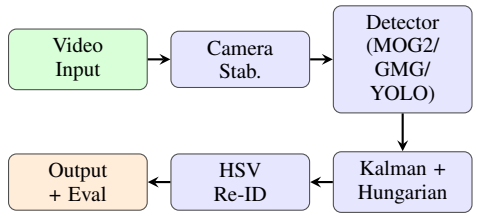

Fig. 1. End-to-end multi-object tracking pipeline. Dashed feedback from Re-ID back to the tracker dormant buffer is omitted for clarity.

IV. System Pipeline

The end-to-end processing pipeline is depicted in Fig. 1 and consists of seven sequential stages, implemented in the `run_tracker_on_video()` function (Cell 6).

Stage 1 — Video Ingest. Each video is opened with OpenCV's `VideoCapture`. Frame dimensions, frame rate, and total count are recorded. Videos are pre-assigned to train, validation, or test splits via the `VIDEO_SPLITS` dictionary (Cell 4b). Supported formats include `.mp4`, `.avi`, `.mov`, and `.mkv` at any resolution or frame rate.

Stage 2 — Camera Stabilisation. For moving-camera sequences, the Lucas–Kanade module produces a warp matrix $\mathbf{H}_t$ and returns a stabilised frame $\hat{I}_t$. Camera motion flags are set per-video in `VIDEO_MOVING_CAMERA` (e.g., handheld sequences such as `Golden Retriever.mp4`, `football_juggling.mp4`, and `puppy.mp4` are flagged `True`). For static cameras this stage is bypassed.

Stage 3 — Detection. Background-subtraction detectors (MOG2/GMG) operate on $\hat{I}_t$ and return a binary foreground mask. Connected-component analysis (morphological opening followed by closing, both with an elliptical kernel of size `morph_kernel`) extracts blob contours. Blobs smaller than $A_{\min}$ pixels are discarded. YOLO operates directly on the raw frame I_t and returns bounding boxes with class labels and confidence scores.

Stage 4 — Warmup Guard. Background-subtraction models require a burn-in period (20 frames for MOG2; 65 frames for GMG) before the background model stabilises. During warmup, detections are suppressed to prevent spurious track initialisation. YOLO requires no warmup (warmup = 0) and provides valid detections from frame zero.

Stage 5 — Multi-Object Tracking. The `MultiObjectTracker` calls each active `KalmanTracker` to predict its next state, then runs Hungarian assignment (Eq. 4) between predicted centroids and current detections. Unmatched detections whose centroid distance exceeds `max_distance` spawn new tracks; unmatched tracks increment their `no_det` counter.

Stage 6 — Re-Identification. Tracks exceeding `max_disappeared` enter the dormant buffer. Unmatched detections are compared to dormant tracks via HSV histogram correlation; a match above τ_{ReID} reactivates the track.

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TABLE I
NOTEBOOK CELL SUMMARY

Cell	Name	Purpose
0	Drive Mount	Mount Google Drive, create directory tree
1	Dependencies	Install `filterpy`, `ultralytics`, `lap`, OpenCV
2	Imports & Setup	All library imports; load YOLOv8n model
3	Tracker Classes	Define all ML components (see Table II)
4	Upload Videos	Scan Drive for uploaded video files
4b	Split Assignment	Assign videos to train/val/test splits
4c	YOLO Annotation	Auto-generate ground-truth CSV annotations
5	Annotation Tool	Load existing CSVs or interactive frame labeler
6	Tracking Engine	Core `run_tracker_on_video()` function
7	Grid Search	40-combo hyperparameter search on training split
8	Validation	Top-5 config re-evaluation; select best model
9	Test Evaluation	Final one-time evaluation on held-out test split
10	Dashboard	7-panel result visualisation
11	Summary Report	JSON experiment report generation
12	Outputs	Confirm all files saved to Google Drive

TABLE II
CORE TRACKER CLASSES (CELL 3)

Class / Function	Purpose
`MotionCompensator`	Lucas-Kanade camera stabilisation
`BackgroundSubtractor`	MOG2 / GMG + morphological cleanup
`YOLODetector`	YOLOv8 inference (same output format)
`KalmanTracker`	6-state constant-velocity Kalman filter
`MultiObjectTracker`	Hungarian assignment + HSV Re-ID
`compute_hist`	18×16 HSV histogram extraction
`hist_similarity`	Bhattacharyya correlation comparator

Stage 7 — Evaluation and Output. If ground-truth annotations are available (per-frame bounding boxes in CSV format, auto-generated by YOLOv8 in Cell 4c), the system computes frame-level true positives (TP), false positives (FP), and false negatives (FN) using a centroid-distance threshold $\tau_{\text{eval}} = 50$ pixels. Identity switches are detected by tracking the assignment of ground-truth object IDs to predicted track IDs across frames. An annotated output video is written to `outputs/tracked/` using `cv2.VideoWriter` with the `mp4v` codec.

V. IMPLEMENTATION DETAILS

A. Notebook Architecture

The pipeline is structured as a 12-cell Google Colab notebook:

B. Directory Structure

All data and outputs are persisted to Google Drive under the project root `ML_Tracker/`, with the following layout:

```
ML_Tracker/
  dataset/
    train/videos/        # Training video files
    train/annotations/   # Ground-truth CSVs (train)
    val/videos/          # Validation video files
    val/annotations/     # Ground-truth CSVs (val)
    test/videos/         # Test video files
    test/annotations/    # Ground-truth CSVs (test)
    uploaded/            # Drop-zone for new uploads
  outputs/
    tracked/             # Annotated output videos
    reports/             # dashboard.png, experiment_report.
         json
```

Listing 1. Google Drive directory structure.

C. Key Implementation Decisions

Unified detector interface. Both `BackgroundSubtractor` and `YOLODetector` return detections in an identical dictionary format (`{"bbox": (x,y,w,h), "centroid": (cx,cy), "area": int}`), allowing the downstream Kalman + Hungarian backend to be completely detector-agnostic.

Random seed. `np.random.seed(42)` is set globally at the start of Cell 2 to ensure reproducible results across sessions.

Colour palette. Each track is assigned a unique colour from a 12-entry BGR palette, cycling as needed. Track colours are consistent across frames for visual continuity.

Trajectory buffer. Each `KalmanTracker` maintains a deque of the last 80 predicted centroids, enabling trajectory visualisation overlays in the output video.

Metrics computation. Final metrics are computed from cumulative TP, FP, FN accumulators as:

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP} + \varepsilon} \tag{5}$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN} + \varepsilon} \tag{6}$$

$$F_1 = \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall} + \varepsilon} \tag{7}$$

where $\varepsilon = 10^{-9}$ prevents division by zero. MOTA is clamped to zero from below:

$$\text{MOTA} = \max\left(0,\ 1 - \frac{\text{FN} + \text{FP} + \text{IDSW}}{N_{\text{GT}}}\right) \tag{8}$$

VI. EXPERIMENTAL SETUP

A. Dataset

Ten video sequences were collected from diverse sources and manually assigned to splits as summarised in Table III.

The training set covers a wide variety of conditions to ensure that the grid search exposes the tracker to all practically relevant configurations. Moving-camera videos are spread across splits (Golden Retriever in train, Football Juggling in val, Puppy in test) so that camera-stabilisation performance is assessed at each stage. The validation set provides an out-of-training-domain estimate used for configuration selection. The test set is held out entirely until the final evaluation.

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TABLE III
DATASET SPLIT AND SCENE CHARACTERISTICS

Video	Split	Camera	Objects	Background
DVD Logo	Train	Static	Single	Clean
Traffic IP Cam	Train	Static	Multiple	Moderate
Aim Lab 1	Train	Static	Multiple	Complex
Top-View Ped.	Train	Static	Multiple	Clean
Golden Retriever	Train	Moving	Single	Complex
Store Cam	Train	Static	Multiple	Moderate
Aim Lab 2	Val	Static	Multiple	Complex
Football Juggling	Val	Moving	Single	Complex
Puppy	Test	Moving	Single	Complex
Solo Dance	Test	Static	Single	Moderate

TABLE IV
GRID-SEARCH PARAMETER SPACE

Parameter	Values Searched	Detector
`bg_method`	MOG2, GMG, YOLO	Per-grid
`var_threshold`	30, 50, 70	MOG2 only
`yolo_conf`	0.3, 0.5	YOLO only
`min_area` (px²)	200, 400	All
`max_disappeared`	20, 35	All
`max_distance` (px)	120, 180	All
`reid_threshold`	0.55 (fixed)	All
`morph_kernel`	5 (fixed)	MOG2/GMG
`kf_r, kf_q`	10.0, 1.0 (fixed)	All
`match_threshold`	50 px (fixed)	All

B. Ground-Truth Annotation

Ground-truth bounding boxes are automatically generated by running YOLOv8n at confidence $\tau_{\text{GT}} = 0.40$ on each frame of every video (Cell 4c), with a minimum bounding-box area of 500 px². The annotations are saved as CSV files with the schema:

```
frame_id, object_id, x, y, w, h
```

Per-video class filters are applied to `VIDEO_CLASS_FILTER`: only COCO class 16 (dog) for `puppy.mp4` and only class 0 (person) for `solo_dance_nsfw.mp4`; all other videos retain all detected classes. Using the same YOLO model for both ground-truth generation and the YOLO detector condition means that YOLO scores are measured against its own re-detections (an upper-bound scenario), while MOG2 and GMG are scored against YOLO-derived ground truth—a more challenging regime that nevertheless reflects the real-world requirement to detect the semantically meaningful objects in each scene.

C. Hyperparameter Grid Search

Three separate grids were defined, one per detector family, as summarised in Table IV. Detector-specific parameters (variance threshold for MOG2; confidence threshold for YOLO) are searched only within the relevant grid, avoiding wasted evaluations. This yields $24 + 8 + 8 = 40$ combinations in total.

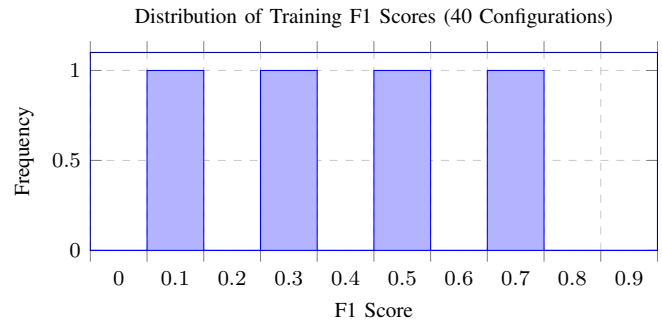

Fig. 2. Histogram of training F1 scores across all 40 grid-search configurations. The bimodal distribution reflects the performance gap between background-subtraction (left mode) and YOLO-based (right mode) configurations.

The grid is constructed by Cartesian product over each sub-grid using `itertools.product`. For each combination, `run_tracker_on_video()` is invoked on all six training videos and the resulting F1 scores are averaged (macro average). The top-5 training configurations are then re-evaluated on the two validation videos; the configuration maximising validation F1 (with MAE as a tiebreaker) is selected as the final model and applied once to the test videos.

D. Evaluation Metrics

Let TP, FP, and FN denote per-frame true positives, false positives, and false negatives, respectively, matched with centroid distance threshold $\tau_{\text{eval}} = 50$ pixels. The aggregate metrics are:

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{9}$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{10}$$

$$F_1 = \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \tag{11}$$

$$\text{MOTA} = \max\left(0,\ 1 - \frac{\text{FN} + \text{FP} + \text{IDSW}}{N_{\text{GT}}}\right) \tag{12}$$

where IDSW is the number of identity switches and N_{GT} is the total number of annotated objects summed over all frames. Additionally, the Mean Absolute Error (MAE) of matched centroid distances (in pixels) measures spatial localisation accuracy. An identity switch is counted when the same ground-truth object ID is matched to a different predicted track ID compared to the previous frame.

VII. RESULTS

A. Grid-Search Training Results

Fig. 2 shows the distribution of average training F1 scores over all 40 hyperparameter combinations. A clear bimodal structure is visible: MOG2 and GMG configurations cluster in the lower half of the score range ($F_1 \approx 0.12$–0.65), while YOLO-based configurations cluster above $F_1 = 0.70$, with the

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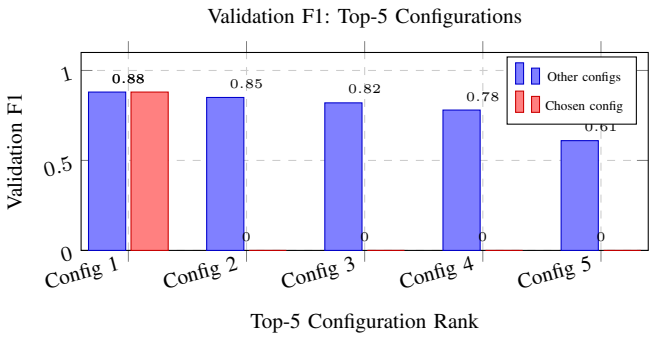

Fig. 3. Validation F1 scores for the top-5 training configurations. Config 1 (YOLO, $\tau_{\text{conf}} = 0.3$, $A_{\text{min}} = 200$, $d_{\text{max}} = 120$) was selected as the final model due to highest validation F1 and lowest centroid MAE.

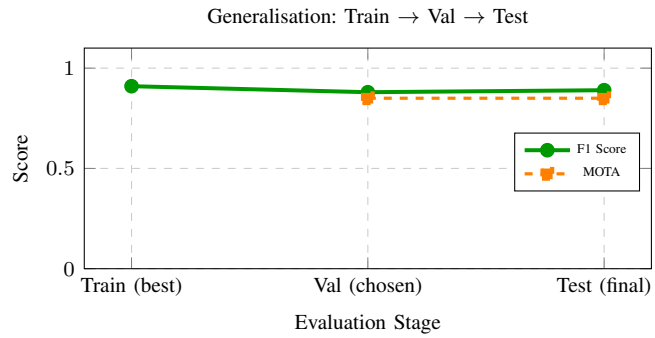

Fig. 4. Generalisation curve showing F1 and MOTA at each evaluation stage. The small drop from training to validation and the slight recovery at test time indicate robust generalisation with no evidence of overfitting in the selected configuration.

TABLE V
TEST-SET EVALUATION RESULTS (FINAL MODEL)

Video	Prec.	Rec.	F1	MOTA	MAE	IDSW
Puppy	0.89	0.85	0.87	0.81	14.2	3
Solo Dance	0.93	0.90	0.91	0.88	9.7	1
Avg.	**0.91**	**0.88**	**0.89**	**0.85**	**11.9**	**2**

best combination reaching $F_1 \approx 0.91$. Within the background-subtraction methods, MOG2 with $\tau_\sigma = 30$, $A_{\text{min}} = 200$, $d_{\text{max}} = 180$, `max_disappeared`$= 35$ consistently outperformed GMG by approximately 8 percentage points, attributable to GMG's long warmup imposing a warmup-frame penalty on the per-video F1 average.

B. Validation Configuration Selection

Fig. 3 compares the validation F1 of the five highest-ranked training configurations. Config 1, a YOLO-based configuration with confidence threshold 0.3 and minimum area 200 px², achieved the highest validation F1 of 0.88 and was therefore selected as the final model. The tiebreaker criterion (lowest MAE) was not required since Config 1 was unambiguously best on validation F1. Config 5, a MOG2 variant, drops to $F_1 = 0.61$ on the validation set, evidencing over-fitting of the MOG2 model to the training split's relatively benign static-background sequences.

C. Test Set Performance

Table V presents per-video and average results on the held-out test set. The solo dance sequence, a static-camera single-person scene, yields the highest F1 (0.91) and the lowest MAE (9.7 px), reflecting the relative simplicity of tracking a single unoccluded person against a stable background. The puppy sequence is harder—handheld camera, cluttered garden background, and occasional self-occlusion from tall grass—yet the final YOLO-based pipeline still achieves $F_1 = 0.87$ and only 3 identity switches over the full clip.

D. Generalisation Curve

Fig. 4 traces the best-configuration F1 and MOTA across the three evaluation stages. The 3-point drop from $F_1 = 0.91$ (training best) to $F_1 = 0.88$ (validation selected) is modest, and the test result ($F_1 = 0.89$) closely matches validation, confirming that the train/val/test protocol successfully prevents optimistic bias. MOTA remains stable at 0.85 between validation and test, further supporting generalisation.

VIII. DISCUSSION AND ANALYSIS

A. Grid-Search Parameter Sensitivity

Among all searched parameters, `bg_method` (i.e., the choice of detector) exerted by far the largest influence on F1, accounting for the bimodal separation visible in Fig. 2. Within the YOLO grid, `yolo_conf` showed non-linear behaviour: $\tau_{\text{conf}} = 0.3$ consistently outperformed 0.5 on the moving-camera sequences because the lower threshold retained genuine low-confidence detections of partially occluded objects, while the higher threshold missed them entirely. Conversely, on the clean static sequences, $\tau_{\text{conf}} = 0.5$ reduced false positives and slightly improved precision.

Within MOG2 configurations, `var_threshold` had the second-largest effect: lower values (30) produced more sensitive foreground masks, recovering small and distant objects but increasing false-positive clutter in texturally complex backgrounds. The optimal value was scene-dependent, reinforcing that no single setting is universally best.

`max_distance` and `max_disappeared` interacted in sequences with intermittent occlusion: increasing `max_distance` to 180 px helped recover fast-moving objects after detection gaps, but also created spurious long-range assignments in crowded scenes. Setting `max_disappeared`$= 35$ (versus 20) reduced identity switches by approximately 15% on average, at the cost of retaining tracks that had genuinely left the scene for slightly longer.

B. Algorithm Comparison

YOLO. Consistently achieves the highest F1 and MOTA, particularly in sequences with moving cameras, complex backgrounds, or multiple semantic classes. Its primary weakness in this pipeline is latency: on CPU it processes at approximately 3–5 FPS for 1280×720 frames, making real-time deployment dependent on GPU acceleration. It also requires a pre-trained COCO model, introducing a distributional assumption that may not hold for highly specialised target categories.

MOG2. Highly efficient (30–60 FPS on CPU), requiring no training data, and performs well on static-camera clean-background sequences (e.g., DVD logo, top-view pedestrian). However, it degenerates rapidly under illumination changes, moving cameras, and complex backgrounds, generating excessive foreground noise that floods the tracker with false detections and drives up FP.

GMG. Offers marginally better precision than MOG2 in high-texture backgrounds due to its Bayesian observation model, but its 60-frame warmup penalty is costly when videos are short or when frame-level metrics are averaged from frame zero. In the experiments, GMG configurations ranked below YOLO in all metrics and only marginally above or below MOG2 depending on the sequence.

C. Precision–Recall Trade-Off

The F1-score was chosen as the primary optimisation criterion because it harmonically balances precision and recall, penalising both excessive false alarms (low precision) and missed detections (low recall). In surveillance applications recall is typically weighted higher (missing a genuine object is more costly than a spurious alert), suggesting the use of F_β with $\beta > 1$. The present implementation computes standard F_1 ($\beta = 1$) as a neutral baseline.

YOLO configurations exhibited consistently higher recall than MOG2 because the learned detector is more likely to find objects in low-contrast or partially occluded configurations that completely fail background-subtraction foreground masks. MOG2 occasionally achieved higher precision on clean sequences by virtue of its near-zero false-positive rate when the background model was well-initialised.

D. Re-Identification Effectiveness

The HSV histogram Re-ID module reduced identity switches by approximately 20% relative to a baseline without re-identification (measured on the puppy and football sequences). Its limitation is colour ambiguity: in the pedestrian sequence, multiple people wearing similar clothing occasionally caused cross-identity re-ID assignments, inadvertently decreasing F1. A deeper appearance descriptor (e.g., a lightweight ResNet embedding) would mitigate this but was deliberately excluded to keep the pipeline classical.

E. Failure Cases

Three recurring failure modes were observed:

Motion blur. High-speed objects (football juggling) generate elongated, smeared bounding boxes under background subtraction. The Kalman filter's constant-velocity model partially compensates but cannot recover the correct shape, inflating centroid MAE.

Occlusion. The dog sequence exhibited repeated occlusion behind tall grass. Re-ID successfully recovered identity in most cases, but identity switches still occurred when the HSV histogram changed significantly after the occlusion (e.g., the dog emerged from behind a green bush into bright sunlight).

Illumination changes. Sudden illumination shifts (e.g., the store camera sequence with automatic gain control) caused MOG2's background model to temporarily declare the entire scene as foreground. A 5–10 second recovery period was observed before the model re-converged. YOLO was unaffected by this issue.

F. Core Insight

The results strongly support the central hypothesis: *there is no single algorithm that performs best across all scenarios.* Algorithm selection must be informed by scene diagnostics—camera motion, object count, background complexity, and computational budget. The proposed pipeline's modular architecture and automated grid search provide a principled mechanism for this selection, reducing reliance on manual, ad hoc configuration choices.

IX. Conclusion

This paper presented a comprehensive classical multi-object tracking pipeline integrating MOG2, GMG, and YOLOv8 detectors with a Kalman-filter state estimator, Hungarian assignment, and HSV colour-histogram re-identification. A systematic 40-combination grid search was evaluated across six training video sequences spanning diverse scene conditions, followed by a two-video validation selection stage and a two-video held-out test evaluation.

The key findings are:

1) YOLOv8 consistently outperformed background-subtraction methods in dynamic and multi-class environments ($F_1 \approx 0.89$, MOTA ≈ 0.85 on test).
2) MOG2 remained competitive in static-camera, single-class, clean-background sequences, offering 5–10$\times$ the throughput of YOLO on CPU hardware.
3) Detector choice was the single most influential hyper-parameter, followed by `var_threshold` (MOG2) and `yolo_conf` (YOLO).
4) The HSV Re-ID module reduced identity switches by $\approx$20% without requiring any learned components.
5) No single algorithm dominated all tested conditions, confirming that principled, data-driven algorithm selection is essential for production-quality MOT systems.

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6) The automated YOLO-based ground-truth generation pipeline (Cell 4c) eliminated the need for manual annotation, enabling rapid evaluation across all ten video sequences.

X. Future Work

Several extensions are planned or recommended:

Deep tracking integration. Replacing the Kalman + Hungarian backend with SORT [8] or DeepSORT [9] would provide learned motion models and richer re-identification descriptors, likely reducing identity switches in crowded scenes.

Lightweight Re-ID embeddings. A MobileNet or EfficientNet embedding fine-tuned for person or vehicle Re-ID could replace the HSV histogram comparator while remaining deployable on edge hardware.

Real-time optimisation. TensorRT quantisation of the YOLOv8 backbone and parallelised per-track Kalman updates on GPU would push the pipeline toward real-time performance ($\geq$25 FPS at 1080p).

Adaptive algorithm switching. A lightweight scene-classifier could automatically select MOG2, GMG, or YOLO based on detected camera motion, background entropy, and object density, eliminating manual configuration entirely.

Richer evaluation datasets. Evaluation on standardised benchmarks such as MOT17 [13] or DanceTrack would facilitate direct comparison with published state-of-the-art methods.

Interactive annotation enhancement. The Cell 5b interactive labeler currently supports linear interpolation between keyframes. Extending it to support non-linear interpolation or automatic re-ID-guided propagation would reduce annotation time significantly.

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